

Mensuration (9 th std)

Choose the Appropriate Answer

1. The semi-perimeter of a triangle having sides 15 cm, 20 cm and 25 cm is

- (a) 60 cm (b) 45 cm (c) 30 cm (d) 15 cm

Ans: (c) 30 cm

$$\text{Semi-perimeter} = (a + b + c) / 2 = (15 + 20 + 25) / 2 = 30 \text{ cm}$$

2. If the sides of a triangle are 3 cm, 4 cm and 5 cm, then the area is

- (a) 3 cm² (b) 6 cm² (c) 9 cm² (d) 12 cm²

Ans: (b) 6 cm²

$$s = (a + b + c) / 2 = (3 + 4 + 5) / 2 = 6 \text{ cm}$$

$$\text{Area of the triangle} = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{6 \times 3 \times 2 \times 1} = \sqrt{36} = 6 \text{ cm}^2$$

3. The perimeter of an equilateral triangle is 30 cm. The area is

- (a) 10 $\sqrt{3}$ cm² (b) 12 $\sqrt{3}$ cm² (c) 15 $\sqrt{3}$ cm² (d) 25 $\sqrt{3}$ cm²

Ans: (d) 25 $\sqrt{3}$ cm²

$$\text{Perimeter of an equilateral triangle} = 30 \text{ cm} \Rightarrow 3a = 30 \text{ cm} \Rightarrow a = 30 / 3 = 10 \text{ cm}$$

$$\text{Area of an equilateral triangle} = \frac{\sqrt{3}}{4} a^2 \text{ sq. Units} = \frac{\sqrt{3}}{4} \times 10 \times 10 = 25 \sqrt{3} \text{ cm}^2$$

4. The lateral surface area of a cube of side 12 cm is

- (a) 144 cm² (b) 196 cm² (c) 576 cm² (d) 664 cm²

Ans: (c) 576 cm²

$$\text{L.S.A. of a cube} = 4a^2 \text{ sq. Units} = 4 \times 12 \times 12 \text{ cm}^2 = 576 \text{ cm}^2$$

5. If the lateral surface area of a cube is 600 cm², then the total surface area is

- (a) 150 cm² (b) 400 cm² (c) 900 cm² (d) 1350 cm²

Ans: (c) 900 cm²

$$\text{L.S.A. of a cube} = 4a^2 = 600 \Rightarrow a^2 = 600 / 4 = 150$$

$$\text{Total surface area of a cube} = 6a^2 \text{ sq. Units} = 6 \times 150 \text{ cm}^2 = 900 \text{ cm}^2$$

6. The total surface area of a cuboid with dimension 10 cm × 6 cm × 5 cm is

- (a) 280 cm² (b) 300 cm² (c) 360 cm² (d) 600 cm²

Ans: (a) 280 cm²

$$\begin{aligned} \text{T.S.A. of a cuboid} &= 2(lb + bh + lh) = 2(10 \times 6 + 6 \times 5 + 10 \times 5) \text{ cm}^2 \\ &= 2(60 + 30 + 50) = 2 \times 140 = 280 \text{ cm}^2 \end{aligned}$$

7. If the ratio of the sides of two cubes are 2: 3, then ratio of their surface areas will be

- (a) 4 : 6 (b) 4 : 9 (c) 6 : 9 (d) 16 : 36

Ans: (b) 4 : 9

$$\text{Ratio of the surface area of cubes} = 4a_1^2 : 4a_2^2 \Rightarrow a_1^2 : a_2^2 = 4^2 : 9^2 = 4 : 9$$

8. The volume of a cuboid is 660 cm and the area of the base is 33 cm². Its height is

- (a) 10 cm (b) 12 cm (c) 20 cm (d) 22 cm

Ans: (c) 20 cm

$$\text{Volume of a cuboid} = l \times b \times h = 660 \Rightarrow 33 \times h = 660 \Rightarrow h = 660 / 33 = 20 \text{ cm}$$

9. The capacity of a water tank of dimensions 10 m × 5 m × 1.5 m is

- (a) 75 litres (b) 750 litres (c) 7500 litres (d) 75000 litres

Ans: (d) 75000 litres

$$\text{The capacity of a tank} = l \times b \times h = (10 \times 5 \times 1.5) \text{ m}^3 = 75 \text{ m}^3 = 75000 \text{ litres}$$

10. The number of bricks each measuring 50 cm × 30 cm × 20 cm that will be required to build a wall whose dimensions are 5 m × 3 m × 2 m is

- (a) 1000 (b) 2000 (c) 3000 (d) 5000

Ans: (a) 1000

$$\begin{aligned} \text{Volume of one brick} &= 50 \times 30 \times 20 \text{ cm}^3, \text{ Volume of the wall} = l \times b \times h = 500 \times 300 \times 200 \text{ cm}^3 \\ \text{No. of bricks} &= \frac{\text{Volume of the wall}}{\text{Volume of one bricks}} = \frac{500 \times 300 \times 200}{50 \times 30 \times 20} = 10 \times 10 \times 10 = 1000 \text{ bricks} \end{aligned}$$

11. If the sides of a triangles are 5 cm, 6 cm and 7 cm then the area is

- (a) 18 cm² (b) 6 √2 cm² (c) 6 √6 cm² (d) 6 √3 cm²

Ans: (c) 6 √6 cm²

12. The perimeter of an equilateral triangle is 60 cm then the area is

- (a) 60 √3 cm² (b) 20 √3 cm² (c) 50 √3 cm² (d) 100 √3 cm²

Ans: (d) 100 √3 cm²

13. The total surface area of the cuboid with dimension 20 cm × 30 cm × 15 cm is ...

- (a) 2700 cm² (b) 1500 cm² (c) 2500 cm² (d) 3000 cm²

Ans: (a) 2700 cm²

14. The number of bricks each measuring 70 cm × 80 cm × 40 cm that will be required to build a wall whose dimensions are 7 m × 8 m × 4 m is

- (a) 4000 (b) 3000 (c) 2000 (d) 1000

Ans: (d) 1000

15. The volume of a cube is 4913 m³ then the length of its side is

- (a) 13 m (b) 17 m (c) 34 m (d) 27 m

Ans: (b) 17 m

16. The formula to find the surface area of a cuboid of length (l), breadth (b) and height (h) is:

- (a) l b + b h + h l (b) 2 (l b + b h + h l) (c) 2 (l b h) (d) l b h / 2

Ans: (b) 2 (l b + b h + h l)

17. The surface area of a cube whose edge equals to 3cm is:
 (a) 62 sq.cm (b) 30 sq.cm (c) 54 sq.cm (d) 90 sq.cm

Ans: (c) 54 sq.cm

$$\text{Surface area of cube} = 6a^2 = 6 \times 3 \times 3 = 54 \text{ sq.cm}$$

18. The surface area of cuboid-shaped box having length = 80 cm, breadth = 40cm and height = 20cm is:

(a) 11200 sq.cm (b) 13000 sq.cm (c) 13400 sq.cm (d) 12000 sq.cm

Ans: (a) 11200 sq.cm

$$\begin{aligned} \text{Surface area of the box} &= 2(lb + bh + hl) = 2[(80 \times 40) + (40 \times 20) + (20 \times 80)] \\ &= 2[3200 + 800 + 1600] = 2 \times 5600 = 11200 \text{ sq.cm.} \end{aligned}$$

19. The lateral surface area of a cube is 256 m². The volume of the cube is

(a) 512 m³ (b) 64 m³ (c) 216 m³ (d) 256 m³

Ans: (a) 512 m³

$$\begin{aligned} \text{The lateral surface area of cube} &= 4a^2 = 256 \Rightarrow a^2 = 256 / 4 = 64 \Rightarrow a = 8 \text{ m} \\ \text{The volume of cube} &= a^3 \text{ cube units} \Rightarrow V = 8^3 = 512 \text{ m}^3 \end{aligned}$$

20. The total surface area of a cube is 96 cm². The volume of the cube is:

(a) 8 cm³ (b) 512 cm³ (c) 64 cm³ (d) 27 cm³

Ans: (c) 64 cm³

$$6a^2 = 96 \text{ cm}^2 \Rightarrow a^2 = 96/6 = 16 \Rightarrow a = 4 \text{ cm} \Rightarrow V = 4^3 = 64 \text{ cm}^3$$

21. The length of the longest pole that can be put in a room of dimensions (10×10× 5) is (in meters)

(a) 15m (b) 16m (c) 10m (d) 12m

Ans: (a) 15 m

$$\text{The length of the longest pole} = \sqrt{[10^2 + 10^2 + 5^2]} = \sqrt{(100 + 100 + 25)} = \sqrt{225} = 15 \text{ m.}$$

22. A cuboid having surface areas of 3 adjacent faces as a, b and c has the volume:

(a) $3\sqrt{a b c}$ (b) $\sqrt{a b c}$ (c) $a b c$ (d) $(a b c)^2$

Ans: (b) $\sqrt{a b c}$

23. Area of a triangle is equal to:

(a) Base x Height (b) 2 (Base x Height)
 (c) $\frac{1}{2}$ (Base x Height) (d) $\frac{1}{2}$ (Base + Height)

Ans: (c) $\frac{1}{2}$ (Base x Height)

24. If the perimeter of an equilateral triangle is 180 cm. Then its area will be:

(a) 900 cm² (b) $900\sqrt{3}$ cm² (c) $300\sqrt{3}$ cm² (d) $600\sqrt{3}$ cm²

Ans: (b) $900\sqrt{3}$ cm²

$$\begin{aligned} 3a &= 180 \quad a = 60 \text{ cm} \Rightarrow \text{Semi-perimeter} = 180 / 2 = 90 \text{ cm} \\ \text{Heron's formula,} \\ A &= \sqrt{s(s-a)(s-b)(s-c)} \Rightarrow (a = b = c = 60 \text{ cm}) \\ A &= \sqrt{[90(90-60)(90-60)(90-60)]} = \sqrt{(90 \times 30 \times 30 \times 30)} = 900\sqrt{3} \text{ cm}^2 \end{aligned}$$

25. The sides of a triangle are 122 m, 22 m and 120 m respectively. The area of the triangle is:

- (a) 1320 sq. m (b) 1300 sq. m (c) 1400 sq. m (d) 1420 sq. m

Ans: (a) 1320 sq. m

$$\text{Semi-perimeter, } s = (122 + 22 + 120)/2 = 132 \text{ m}$$

$$\text{Using heron's formula: } A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{[132(132 - 122)(132 - 22)(132 - 120)]} = \sqrt{(132 \times 10 \times 110 \times 12)} = 1320 \text{ sq.m}$$

26. The area of a triangle with given two sides 18 cm and 10 cm, respectively and a perimeter equal to 42 cm is:

- (a) $20\sqrt{11} \text{ cm}^2$ (b) $19\sqrt{11} \text{ cm}^2$ (c) $22\sqrt{11} \text{ cm}^2$ (d) $21\sqrt{11} \text{ cm}^2$

Ans: (d) $21\sqrt{11} \text{ cm}^2$

$$a + b + c = 42 \Rightarrow 18 + 10 + c = 42 \Rightarrow c = 42 - 28 = 14 \text{ cm}$$

$$\text{Semi perimeter, } s = 42/2 = 21 \text{ cm}$$

$$\text{Using Heron's formula: } A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{[21(21 - 18)(21 - 10)(21 - 14)]} = \sqrt{(21 \times 3 \times 11 \times 7)} = 21\sqrt{11} \text{ cm}^2$$

27. The equal sides of the isosceles triangle are 12 cm, and the perimeter is 30 cm. The area of this triangle is:

- (a) $9\sqrt{15} \text{ sq.cm}$ (b) $6\sqrt{15} \text{ sq.cm}$ (c) $3\sqrt{15} \text{ sq.cm}$ (d) $\sqrt{15} \text{ sq.cm}$

Ans: (a) $9\sqrt{15} \text{ sq.cm}$

$$\text{Perimeter} = 30 \text{ cm} \Rightarrow \text{Semi perimeter, } s = 30 / 2 = 15 \text{ cm}$$

$$a + b + c = 30 \Rightarrow 12 + 12 + c = 30 \Rightarrow c = 30 - 24 = 6 \text{ cm}$$

$$\text{Using Heron's formula: } A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{[15(15 - 12)(15 - 12)(15 - 6)]} = \sqrt{(15 \times 3 \times 3 \times 9)} = 9\sqrt{15} \text{ sq.cm}$$

28. The area of an equilateral triangle having side length equal to $\sqrt{3} / 4$ cm (using Heron's formula) is:

- (a) $2 / 27 \text{ sq.cm}$ (b) $2 / 15 \text{ sq.cm}$ (c) $3\sqrt{3} / 64 \text{ sq.cm}$ (d) $3 / 14 \text{ sq.cm}$

Ans: (c) $3\sqrt{3} / 64 \text{ sq.cm}$

$$a = b = c = \sqrt{3} / 4 \Rightarrow \text{Semi perimeter} = (a + b + c) / 2 = 3a / 2 = 3\sqrt{3} / 8 \text{ cm}$$

$$A = \sqrt{s(s-a)(s-b)(s-c)} \Rightarrow A = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{[(3\sqrt{3} / 8)(3\sqrt{3} / 8 - \sqrt{3} / 4)(3\sqrt{3} / 8 - \sqrt{3} / 4)(3\sqrt{3} / 8 - \sqrt{3} / 4)]} = 3\sqrt{3} / 64 \text{ sq.cm}$$

29. The base of a right triangle is 8 cm and the hypotenuse is 10 cm. Its area will be

- (a) 24 cm^2 (b) 40 cm^2 (c) 48 cm^2 (d) 80 cm^2

Ans: (a) 24 cm^2

$$\text{Base} = 8 \text{ cm and Hypotenuse} = 10 \text{ cm} \Rightarrow \text{height} = \sqrt{(10^2 - 8^2)} = \sqrt{36} = 6 \text{ cm}$$

$$\text{area} = (\frac{1}{2}) \times b \times h = (\frac{1}{2}) \times 8 \times 6 = 24 \text{ cm}^2.$$

30. The edges of a triangular board are 6 cm, 8 cm and 10 cm. The cost of painting it at the rate of 9 paisa per cm^2 is

- (a) Rs.2.00 (b) Rs.2.16 (c) Rs.2.48 (d) Rs.3.00

Ans: (b) Rs.2.16

$$s = (6 + 8 + 10)/2 = 12 \text{ cm}$$

$$A = \sqrt{[12(12-6)(12-8)(12-10)]} = \sqrt{[(12)(6)(4)(2)]} = \sqrt{576} = 24 \text{ cm}^2$$

The cost of painting at a rate of 9 paisa per $\text{cm}^2 = 24 \times 9 \text{ paisa} = \text{Rs.2.16}$

31. An isosceles right triangle has an area of 8 cm^2 . The length of its hypotenuse is

- (a) $\sqrt{32} \text{ cm}$ (b) $\sqrt{16} \text{ cm}$ (c) $\sqrt{48} \text{ cm}$ (d) $\sqrt{24} \text{ cm}$

Ans: (a) $\sqrt{32} \text{ cm}$

32. The area of an isosceles triangle having a base 2 cm and the length of one of the equal sides 4 cm, is

- (a) $\sqrt{15} \text{ cm}^2$ (b) $\sqrt{(15/2)} \text{ cm}^2$ (c) $2\sqrt{15} \text{ cm}^2$ (d) $4\sqrt{15} \text{ cm}^2$

Ans: (a) $\sqrt{15} \text{ cm}^2$

33. The perimeter of an equilateral triangle is 60 m. The area is

- (a) $10\sqrt{3} \text{ m}^2$ (b) $15\sqrt{3} \text{ m}^2$ (c) $20\sqrt{3} \text{ m}^2$ (d) $100\sqrt{3} \text{ m}^2$

Ans: (d) $100\sqrt{3} \text{ m}^2$

34. The sides of a triangle are 35 cm, 54 cm and 61 cm, respectively. The length of its longest altitude

- (a) $16\sqrt{5} \text{ cm}$ (b) $10\sqrt{5} \text{ cm}$ (c) $24\sqrt{5} \text{ cm}$ (d) 18 cm

Ans: (c) $24\sqrt{5} \text{ cm}$

35. The sides of a triangle are 56 cm, 60 cm and 52 cm long. Then the area of the triangle is

- (a) 1322 cm^2 (b) 1311 cm^2 (c) 1344 cm^2 (d) 1392 cm^2

Ans: (c) 1344 cm^2

$$s = (56+60+52) / 2 = 84 \text{ cm.}$$

Heron's formula, $A = \sqrt{[s(s-a)(s-b)(s-c)]}$ square units

$$A = \sqrt{[84(84-56)(84-60)(84-52)]} = \sqrt{(1806336)} = 1344 \text{ cm}^2$$

36. If the area of an equilateral triangle is $16\sqrt{3} \text{ cm}^2$, then perimeter of the triangle is

- (a) 48 cm (b) 24 cm (c) 12 cm (d) 36 cm

Ans: (b) 24 cm

$$(\sqrt{3} / 4) a^2 = 16\sqrt{3} \Rightarrow a^2 = [(16\sqrt{3})(4)] / \sqrt{3} \Rightarrow a^2 = 64$$

$$a = 8 \text{ cm} \Rightarrow \text{Perimeter} = 3(8) = 24 \text{ cm.}$$

37. The area of an equilateral triangle with sides $2\sqrt{3} \text{ cm}$ is

- (a) 5.196 cm^2 (b) 0.866 cm^2 (c) 3.496 cm^2 (d) 1.732 cm^2

Ans: (a) 5.196 cm^2

$$A = (\sqrt{3} / 4) (2\sqrt{3})^2 = (\sqrt{3} / 4) (12) = 3\sqrt{3} = 3(1.732) = 5.196 \text{ cm}^2$$

38. The length of each side of an equilateral triangle having an area of $9\sqrt{3} \text{ cm}^2$ is
 (a) 8 cm (b) 36 cm (c) 4 cm (d) 6 cm

Ans: (d) 6 cm

$$\left(\frac{\sqrt{3}}{4}\right) a^2 = 9\sqrt{3} \Rightarrow a^2 = \frac{[(9\sqrt{3})(4)]}{\sqrt{3}} \Rightarrow a^2 = 36 \Rightarrow a = 6 \text{ cm.}$$

39. Total surface area of the container is 6450 cm^2 . A container which contains colour has a capacity to paint 1075 cm^2 . How much containers are required to paint the box completely?

(a) 6 (b) 5 (c) 4 (d) 3

Ans: (a) 6

$$\text{No of containers requires are} = \frac{6450}{1075} = 6$$

40. A water tank of dimensions $2\text{m} \times 3\text{m} \times 1.5\text{m}$ is to be painted leaving its base. Find the total surface area of the tank?

(a) 28 m^2 (b) 27 m^2 (c) 26 m^2 (d) 25 m^2

Ans: (b) 27 m^2

$$\text{The total area of the tank} = 2\{(2 \times 3) + (3 \times 1.5) + (2 \times 1.5)\} = 2\{13.5\} = 27 \text{ m}^2$$

41. A man has built a cubical water tank in his house. All sides of it are equal to 1.8m . Find the total surface area?

(a) 19.48 m^2 (b) 19.44 m^2 (c) 19.45 m^2 (d) 19.46 m^2

Ans: (b) 19.44 m^2

$$\text{Total surface area of the cubical tank} = 6l^2 = 6(1.8)^2 = 19.44 \text{ m}^2$$

42. Write the lateral surface area of a cuboid having length l units, breadth b units and height h units.

(a) $2(l + b)h$ (b) $2(l + h)b$ (c) $3(l + b)h$ (d) $2(h + b)l$

Ans: (a) $2(l + b)h$

43. The surface area of a cube of edge a is

(a) $4a^2$ (b) $6a^2$ (c) $3a^2$ (d) a^2

Ans: (b) $6a^2$

44. The volume of a cuboid of length l , breadth b and height h is

(a) $l b h$ (b) $l b + b h + h l$ (c) $2(l b + b h + h l)$ (d) $2(l + b)h$

Ans: (a) $l b h$

45. The volume of a cube of edge a is

(a) a^2 (b) a^3 (c) a^4 (d) $6a^2$

Ans: (b) a^3

46. If the height of a cuboid becomes zero, it will take the shape of a

(a) Cube (b) Parallelogram (c) Circle (d) Rectangle

Ans: (d) Rectangle

47. The volume of a room is 80 m^3 . The area of the floor is 20 m^2 . The height of the room is

- (a) 1 m (b) 2 m (c) 3 m (d) 4 m

Ans: (d) 4 m

$$V = 80 \text{ m}^3 \Rightarrow \text{area} = 20 \text{ m}^2 \Rightarrow \text{Height} = 80/20 = 4 \text{ m.}$$

48. The floor of a room is a square of side 6 m. Its height is 4 m. The volume of the room is

- (a) 140 m^3 (b) 142 m^3 (c) 144 m^3 (d) 145 m^3

Ans: (c) 144 m^3

$$\text{Volume} = 6 \times 6 \times 4 = 144 \text{ m}^3$$

49. If the edge of a cube is 1 cm then which of the following is its volume?

- (a) 6 m^3 (b) 3 m^3 (c) 1 m^3 (d) None of these

Ans: (c) 1 m^3

50. If the edge of a cube is 1 cm then which of the following is its total surface area?

- (a) 1 cm^2 (b) 4 cm^2 (c) 6 cm^2 (d) None of these

Ans: (c) 6 cm^2

51. If base area of a room 12 m^2 and height is 3 m then its volume is:

- (a) 4 m^3 (b) 36 m^3 (c) 12 m^3 (d) 18 m^3

Ans: (b) 36 m^3

52. Two identical cubes each of total surface area of 6 cm^2 are joined end to end. Which of the following is the total surface area of the cuboid so formed?

- (a) 12 cm^2 (b) 18 cm^2 (c) 10 cm^2 (d) 8 cm^2

Ans: (c) 10 cm^2

53. A cuboid has _____ pairs of identical faces.

- (a) 2 (b) 3 (c) 4 (d) 5

Ans: (b) 3

54. All six faces of a cube are:

- (a) Identical (b) Different (c) Circular (d) Rectangular

Ans: (a) Identical

55. A cylindrical box has _____ curved surface and _____ circular faces, which are identical.

- (a) One, One (b) One, two . (c) two, one (d) two, two

Ans: (b) One, Two

56. A cube has _____ faces.

- (a) 4 (b) 5 (c) 6 (d) 8

Ans: (c) 6

57. Volume of a cuboid of length (l), width (w) and height (h) is:

- (a) $l b h$ (b) $l b + b h + h l$ (c) $2 (l b + b h + h l)$ (d) $2 (l + b) h$

Ans: (a) $l b h$

58. The volume of the cube is:

- (a) β (b) a^3 (c) $\pi r^2 h$ (d) $\frac{1}{3} a^2 h$

Ans: (b) a^3

59. The lateral surface area (LSA) of the cube is:

- (a) a^2 (b) $2a^2$ (c) $3a^2$ (d) $4a^2$

Ans: (d) $4a^2$

60. The total surface area (TSA) of the cube is:

- (a) $6a^2$ (b) $5a^2$ (c) $4a^2$ (d) a^2

Ans: (a) $6a^2$

61. The volume of the cuboid is:

- (a) $a^2 h$ (b) $a h^2$ (c) $\frac{1}{3} a^2 h$ (d) $\frac{2}{3} a^2 h$

Ans: (a) $a^2 h$

62. The lateral surface area (LSA) of the cuboid is:

- (a) ah (b) $2ah$ (c) $3ah$ (d) $4ah$

Ans: (b) $2ah$

63. The total surface area (TSA) of the cuboid is:

- (a) $6ah$ (b) $5ah$ (c) $4ah$ (d) ah

Ans: (c) $4ah$

64. The base area of the cuboid is:

- (a) a^2 (b) h^2 (c) $a^2 h$ (d) ah^2

Ans: (a) a^2

65. The length, breadth and height of a cuboidal solid is 4 cm, 3 cm and 2 cm respectively. Its volume is

- (a) $(4 + 3 + 2) \text{ cm}^3$ (b) $2(4 + 3 + 2) \text{ cm}^3$
(c) $(4 \times 3 \times 2) \text{ cm}^3$ (d) $2(4 + 3) \times 2 \text{ cm}^3$

Ans: (c) $(4 \times 3 \times 2) \text{ cm}^3$

66. The volume of a cuboidal solid of length 8 m and breadth 5 m is 200 m^3 . Find its height.

- (a) 5 m (b) 6 m (c) 15 m (d) 18 m

Ans: (a) 5 m

$$h = \frac{V}{l \times b} \Rightarrow h = \frac{200}{8 \times 5} = \frac{200}{40} = 5 \text{ m}$$

67. Find the semi perimeter of the triangle whose sides are 42 cm, 34 cm and 20 cm.

- (a) 45 cm (b) 36 cm (c) 48 cm (d) None of these

Ans: (c) 48 cm

$$s = \frac{42 + 34 + 20}{2} = 48.$$

68. The surface area of a cuboid is

- (a) $2(lb + bh + lh)$ (b) $3(lb + bh + lh)$
 (c) $2(lb - bh - lh)$ (d) $3(lb - bh - lh)$

Ans: (a) $2(lb + bh + lh)$

69. The surface area of a cube if edge 'a' is

- (a) $7a^2$ (b) $6a^2$ (c) $5a^3$ (d) $5a^2$

Ans: (b) $6a^2$

70. The length, breadth and height of a room is 5m, 4m and 3m. The cost of white washing its four walls at the rate of Rs.7.50 per m^2 is

- (a) Rs.410 (b) Rs.409 (c) Rs.420 (d) Rs.405

Ans: (d) Rs.405

$$\text{L.S.A} = 2h(l + b) = 2 \times 3(5 + 4) = 6 \times 9 = 54 \text{ m}^2 \\ \text{Cost} = 54 \times 7.50 = \text{Rs. } 405.$$

71. The perimeter of floor of rectangular hall is 250 m. The cost of the white washing its four walls is Rs.15000. The height of the room is

- (a) 5m (b) 4m (c) 6m (d) 8m

Ans: (c) 6m

$$\text{Area of 4 walls} = \frac{15000}{10} = 1500 \text{ m}^2 \\ \text{Height} = \frac{\text{Area}}{\text{Perimeter}} = \frac{1500}{250} = 6 \text{ m.}$$

72. The volume of cubic room is 512 d m^3 . Find its one side.

- (a) 7 d m (b) 8 d m (c) 4 d m (d) 10 d m

Ans: (b) 8 d m

$$V = a^3 = 512 \Rightarrow a = \sqrt[3]{512} = 8 \text{ d m}$$

73. The area of three adjacent faces of a cube is x, y and z. Its volume V is

- (a) $V = xyz$ (b) $V^3 = xyz$ (c) $V^2 = xyz$ (d) None of these

Ans: (c) $V^2 = xyz$

74. Two cubes each of edge 12 cm are joined. The surface area of new cuboid is

- (a) 140 cm^2 (b) 1440 cm^2 (c) 144 cm^2 (d) 72 cm

Ans: (b) 1440 cm^2

$$S = 2(24 \cdot 12 + 12 \cdot 12 + 12 \cdot 24) \\ = 2(288 + 144 + 288) = 2(720) = 1440 \text{ cm}^2$$

75. The dimensions of a cuboid are 50 cm x 40 cm x 10 cm. Its volume in liters is:
 (a) 10 liters (b) 12 liters (c) 20 liters (d) 25 liters

Ans: (c) 20 liters

$$V = 50 \text{ cm} \times 40 \text{ cm} \times 10 \text{ cm} = 20,000 \text{ cm}^3$$

$$\text{Volume in liters} = \frac{20,000}{1,000} = 20 \text{ liters}$$

76. The volume of a cuboidal tank is 250 m^3 . If its base area is 50 m^2 then depth of the tank is

(a) 5 m (b) 200 m (c) 300 m (d) 12500 m

Ans: (a) 5 m

$$V = A \times d \Rightarrow 250 = 50 \times d \Rightarrow d = \frac{250}{50} = 5 \text{ m}$$

77. The length, breadth and height of a cuboidal solid is 4 cm, 3 cm and 2 cm respectively. Its volume is

(a) 27 cm^3 (b) 24 cm^3 (c) 48 cm^3 (d) 28 cm^3

Ans: (b) 24 cm^3

78. The volume of a cuboidal solid of length 8 m and breadth 5 m is 280 m^3 . Find its height.

(a) 5 m (b) 6 m (c) 7 m (d) 8 m

Ans: (c) 7 m

79. The lateral surface area of a cuboid is

(a) $2(l \times b \times h)$ (b) $3(lb + bh + lh)$ (c) $2h(l + b)$ (d) $l \times b \times h$

Ans: (c) $2h(l + b)$

80. The volume of a cube if edge 'a' is

(a) $4a^2$ (b) $6a^2$ (c) a^3 (d) $2a^2$

Ans: (c) a^3

81. The length, breadth and height of a room is 6m, 4m and 5 m. find its lateral surface area.

(a) 110 m^2 (b) 109 m^2 (c) 100 m^2 (d) 105 m^2

Ans: (c) 100 m^2

$$LSA = 2h(l + b) = 2(5)(6 + 4) = 10(10) = 100 \text{ m}^2$$

82. The perimeter of floor of squared hall is 160m. Find its area.

(a) 1500 m^2 (b) 1400 m^2 (c) 1600 m^2 (d) 1800 m^2

Ans: (c) 1600 m^2

$$160 \text{ m} = 4 \times \text{side} \Rightarrow \text{side} = \frac{160 \text{ m}}{4} = 40 \text{ m}$$

$$A = \text{side}^2 = 1600 \text{ m}^2$$

83. The volume of rectangular room is 64 dm^3 . Find its dimensions.

- (a) 16 d m, 8 d m, 4 d m (b) 12 d m, 8 d m, 2 d m
(c) 8 d m, 4 d m, 2 d m (d) 10 d m, 15 d m, 20 d m

Ans: (c) 8 d m, 4 d m, 2 d m

84. A matchbox measures $4 \text{ cm} \times 2.5 \text{ cm} \times 1.5 \text{ cm}$. The volume of packet containing 12 such boxes is:

- (a) 160 cm^3 (b) 180 cm^3 (c) 160 cm^2 (d) 180 cm^2

Ans: (b) 180 cm^3

The volume of each matchbox = $l \times b \times h = 4 \times 2.5 \times 1.5 = 15 \text{ cm}^3$ Volume of 12 matchboxes = $12 \times 15 \text{ cm}^3 = 180 \text{ cm}^3$
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85. A cuboidal water tank is 6 m long, 5 m wide and 4.5 m deep. How many liter of water can it hold?

- (a) 1350 liters (b) 13500 liters (c) 135000 liters (d) 135 liters

Ans: (c) 135000 liters

Volume of the cuboidal tank = $l \times b \times h = 6 \times 5 \times 4.5 = 135 \text{ m}^3$ $V = 135 \times 1000 \text{ L} = 135000 \text{ L} \text{ (} 1 \text{ m}^3 = 1000 \text{ L)}$
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86. A cuboidal vessel is 10 m long and 8 m wide. How high must it be made to hold 380 cubic meters of a liquid?

- (a) 4.75 m (b) 7.85 m (c) 4.75 cm (d) None of these

Ans: (a) 4.75 m

$h = \frac{380}{80} = 4.75 \text{ m}$

87. The capacity of a cuboidal tank is 50000 liters. The length and depth are respectively 2.5 m and 10 m. Its breadth is

- (a) 4 m (b) 3 m (c) 2 m (d) 5 m

Ans: (c) 2 m

Volume = Length $\times$ Breadth $\times$ Depth. ($50,000 \text{ liters} = 50 \text{ m}^3$) $50 = 2.5 \times \text{Breadth} \times 10 \implies 50 = 25 \times \text{Breadth}$ $\implies \text{Breadth} = \frac{50}{25} = 2 \text{ m}$

88. A godown measures $40 \text{ m} \times 50 \text{ m} \times 30 \text{ m}$. Find the maximum number of wooden crates each measuring $15 \text{ m} \times 25 \text{ m} \times 20 \text{ m}$ that can be stored in the godown.

- (a) 18000 (b) 16000 (c) 15000 (d) 14000

Ans:

$n = \frac{V_g}{V_c} = \frac{40 \text{ m} \times 50 \text{ m} \times 30 \text{ m}}{15 \text{ m} \times 25 \text{ m} \times 20 \text{ m}} = \frac{60,000 \text{ m}^3}{7,500 \text{ m}^3} = 8$
--

89. A cube has a volume of 125 cm^3 . What is the length of its side?

- (a) 4 cm (b) 6 cm (c) 5 cm (d) 7 cm

Ans: (c) 5 cm

90. If the length of a cuboid is doubled, breadth is halved, and height remains same, its volume will:

- (a) Remain the same (b) Become double
(c) Become half (d) Become four times

Ans: (a) Remain the same

91. A cuboid has dimensions $8 \text{ cm} \times 6 \text{ cm} \times 5 \text{ cm}$. Its total surface area is:

- (a) 118 cm^2 (b) 236 cm^2 (c) 240 cm^2 (d) 480 cm^2

Ans: (b) 236 cm^2

$$TSA = 2(8 \times 6 + 6 \times 5 + 5 \times 8) = 2(48 + 30 + 40) = 2(118) = 236 \text{ cm}^2$$

92. Find the ratio of the total surface area and lateral surface area of the cube.

- (a) 1: 2 (b) 2: 3 (c) 3: 2 (d) 2: 1

Ans: (c) 3: 2

$$\text{Ratio of TSA and LSA} = 6a^2 / 4a^2 = 3 / 2 = 3: 2$$

93. Volume of a cuboid is 1176 cm^3 , its length and breadth are 12 cm and 7 cm respectively. What is the height of the cuboid?

- (a) 24 cm (b) 11 cm (c) 14 cm (d) 20 cm

Ans: (c) 14 cm

$$\text{Volume of a cuboid} = l * b * h \Rightarrow 12 * 7 * h = 1176 \Rightarrow h = 14 \text{ cm.}$$

94. Volume of a cube of sides equal to l is given by _____

- (a) $l * b * h$ (b) l^3 (c) l^2 (d) $\pi * l * b * h$

Ans: (b) l^3

95. A wall of dimensions $3 \text{ m} * 12 \text{ cm} * 7 \text{ m}$ is to be made by using bricks of dimensions $25 \text{ cm} * 12 \text{ cm} * 10 \text{ cm}$. How much bricks are required to construct the wall?

- (a) 825 (b) 820 (c) 850 (d) 840

Ans: (d) 840

$$\text{Volume of the wall} = 300 * 12 * 700 = 2520000 \text{ cm}^3$$

$$\text{Volume of bricks} = 25 * 12 * 10 = 3000 \text{ cm}^3$$

$$\text{No of bricks required} = \frac{\text{Volume of the wall}}{\text{Volume of one brick}} = \frac{252000}{3000} = 840$$

96. What is the cost of digging a pit of cuboidal shape dimensions of which are $5 \text{ m} * 6 \text{ m} * 3 \text{ m}$ at the rate of $\text{₹}35 / \text{m}^3$?

- (a) $\text{₹}3000$ (b) $\text{₹}3150$ (c) $\text{₹}3375$ (d) $\text{₹}3225$

Ans: (b) $\text{₹}3150$

$$\text{Volume of the pit} = 5 * 6 * 3 = 90 \text{ m}^3, \text{ Cost of } 1 \text{ m}^3 = \text{₹}35$$

$$\text{Cost of digging } 90 \text{ m}^3 = 90 * 35 = \text{₹}3150.$$

97. The surface area of a cube is 1374 square centimeter. Find its length: (Approx.)
 (a) 14.12 cm (b) 15. 13 cm (c) 16. 2 cm (d) 16. 45 cm

Ans: (b) 15. 13 cm

$$6s^2 = 1,374 \Rightarrow s^2 = \frac{1,374}{6} = 229 \Rightarrow s = \sqrt{229} \approx 15.1327...$$

98. Two cubes each of edge 12 cm are joined. The surface area of new cuboid is
 (a) 140 cm² (b) 1440 cm² (c) 144 cm² (d) 72 cm²

Ans: (b) 1440 cm²

$$\begin{aligned} TSA &= 2(24 \times 12 + 12 \times 12 + 12 \times 24) \\ &= 2(288 + 144 + 288) = 2(720) = 1440 \text{ cm}^2 \end{aligned}$$

99. The difference between the total surface area and lateral surface area of a cube of side 4cm is -----

- (a) 28 cm² (b) 30 cm² (c) 32 cm² (d) 34 cm²

Ans: (c) 32 cm²

$$Difference = 6a^2 - 4a^2 = 2a^2 = 2(4^2) = 2(16) = 32 \text{ cm}^2$$

100. If one side is 1 cm. Each side of a cube is increased by 10%. What is the new surface area of the cube?

- (a) 6 cm² (b) 6.23 cm² (c) 7.26 cm² (d) 7.24 cm²

Ans: (c) 7.26 cm²

$$\begin{aligned} SA_1 &= 6 \times (1 \text{ cm})^2 = 6 \text{ cm}^2 \Rightarrow s_2 = 1 \text{ cm} + (10\% \text{ of } 1 \text{ cm}) = 1 \times 1.1 = 1.1 \text{ cm} \\ SA_2 &= 6 \times (1.1 \text{ cm})^2 = 6 \times 1.21 \text{ cm}^2 = 7.26 \text{ cm}^2. \end{aligned}$$